

Mohammad Taha Khan

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Education

Military Institute of Science and Technology

Apr 2021 – Jul 2025

B.Sc. in Nuclear Engineering

CGPA: 3.58/4.0 (3.87 in final year)

Coursework: Neutronics, Reactor Physics, Numerical Methods, Radiation Detection, Radio-chemistry

Research Experience

Research Assistant

Jan 2026 – Present

Department of Nuclear Science and Engineering, Military Institute of Science and Technology

Undergraduate Research Work (Thesis)

Apr 2024 – Jul 2025

- BSc thesis title: A Study on Core Neutronic Parameters of Thermal (BEAVRS PWR) and Fast (NEA SFR) Reactors using Monte Carlo Codes
- Research supervisor: Dr. Abdus Sattar Mollah, Professor of Nuclear Science and Engineering, Military Institute of Science and Technology

Research Interest

- Monte Carlo Transport Simulation
- Digital Twin
- Physics-informed Machine Learning
- Nuclear Space Propulsion
- Neutronics and CFD Coupling
- Deep Learning

Project

Neutronic Analysis of Sodium Cooled Fast Reactor

github.com/tahakhan33/SFR-MOX-3600

- Developed a comprehensive model consisting of hexagonal sub-assemblies, control rods, and radial reflectors, enabling accurate core simulation
- Validated neutronic parameters with OECD/NEA SFR benchmark, confirming valid use of OpenMC
- **Language:** Python; **Libraries:** OpenMC, NumPy, Matplotlib, Math

Publications and Conference Presentation

International Conference on Energy and Power (ICEP) 2022

Dhaka, Bangladesh

- G. R. Khan*, **M. T. Khan**, Neutronic Calculation of Fuel Element of the Modular High Temperature Gas-Cooled Reactor Benchmark Core Design with OpenMC Code

National Conference on Physics 2025

Rajshahi, Bangladesh

- **M. T. Khan***, S. U. K. Sadia, A. S. Mollah, Design Analysis of European Sodium-cooled Fast Reactor Oxide Core (MOX-3600) using OpenMC

- S. U. K. Sadia*, **M. T. Khan**, A. S. Mollah, Neutronic analysis of Benchmark for Evaluation And Validation of Reactor Simulations (BEAVRS) Pressurized Water Reactor Core using OpenMC, *under review in MIJST*

Skills and Certifications

- **Languages:** C, C++, Python, Bash
- **Tools:** MATLAB, Git, SolidWorks, PCTTRAN
- **Documentations:** LaTeX, Markdown
- **Libraries:** OpenMC, NumPy, Matplotlib
- **Platforms:** Windows, Linux
- **Certification:** Certified SolidWorks Associate (CSWA), Sep 2024

Language Proficiency

English: IELTS Band Score 7.5 (Listening: 8.5, Reading: 8.0, Writing: 7.0, Speaking: 6.5)

Bangla: Native

Leadership/Team Work

Department of Nuclear Science and Engineering

Apr 2024 – Jun 2025

Department Captain

- Helped arranging interactive session on "Roadmap to Higher Studies, Career Development and Job Opportunities at Home and Abroad" in association with MIST Nuclear Engineering Club (MNEC) attended by 120+ students
- Worked on updating core policies of MIST with the authority

Additional Experience

Industrial Training

Siddhirganj Combined Cycle Power Plant	06 Mar 2024
Atomic Energy Research Establishment, Center for Research Reactor	05 Mar 2024
Bangladesh Institute of Nuclear Agriculture (BINA)	03 – 04 Mar 2024
Non-Destructive Testing (NDT) Division, Atomic Energy Centre	11 – 15 Feb 2024
Rooppur Nuclear Power Plant (RNPP)	07 – 08 Feb 2024

Volunteering

Volunteer, International Conference on Energy and Power (ICEP) 2022: Coordinated airport pickup and transportation for foreign professors between hotels and conference venue